Class 12 Mathematics

Chapter 3: Matrices

School Worksheet: step-wise solutions

20 questions **49** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 MCQ — 1 mark

Let $A = \begin{bmatrix} 1 & -2 & -1 \\ 0 & 4 & -1 \\ -3 & 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} -2 \\ -5 \\ -7 \end{bmatrix}$, $C = \begin{bmatrix} 9 & 8 & 7 \end{bmatrix}$. Which of the following is defined?

(A) **Only AB** **(B)** Only AC

(C) Only BA **(D)** All AB, AC and BA

Formula / idea: PQ is defined only when (columns of P) $=$ (rows of Q).

1 Write the orders.
$A_{3\times3}, \quad B_{3\times1}, \quad C_{1\times3}$

2 Check AB: columns of A = 3 = rows of B. ✓ 1
AB is defined, of order 3×1

3 Check AC: columns of A = 3, rows of C = 1.
$3 \neq 1 \Rightarrow AC$ is not defined

4 Check BA: columns of B = 1, rows of A = 3.
$1 \neq 3 \Rightarrow BA$ is not defined

5 Hence only AB is defined. Option (A).

Answer **(A) Only AB**

Watch out: Checking rows of the first matrix against columns of the second — it is the other way round.

2 MCQ 1 mark

If $\begin{bmatrix} x+y & 3y \\ 3x & x+3 \end{bmatrix} = \begin{bmatrix} 9 & 4x+y \\ x+6 & y \end{bmatrix}$, then $(x-y)$ is equal to:

(A) -7 **(B)** -3 **(C)** 3 **(D)** 7

Formula / idea: Equal matrices have equal corresponding elements.

1 Equate the $(2,1)$ elements.
$3x = x+6 \Rightarrow x = 3$

2 Equate the $(2,2)$ elements.
$x+3 = y \Rightarrow y = 6$

3 Check with $(1,1)$ and $(1,2)$ elements.
$x+y = 9$✓, $3y = 18 = 4x+y$✓

4 Find $x-y$. ✓ 1
$x-y = 3-6 = -3$

5 Hence option (B).

Answer **(B)** -3

3 MCQ 1 mark

If $A = [a_{ij}]$ is a skew-symmetric matrix of order n, then a_{ii} is equal to:

(A) 1 **(B)** 0 **(C)** -1 **(D)** n

Formula / idea: Skew-symmetric means $A' = -A$, i.e. $a_{ji} = -a_{ij}$ for all i, j.

1 Use the definition of skew-symmetric.
$a_{ji} = -a_{ij}$ for all i, j

2 Put $j = i$ (diagonal element). ✓ 1
$a_{ii} = -a_{ii} \Rightarrow 2a_{ii} = 0 \Rightarrow a_{ii} = 0$

3 Every diagonal element is 0. Option (B).

Answer **(B)** 0

4 MCQ 1 mark

Let $A = [a_{ij}]$ be a 2×2 matrix whose elements are given by $a_{ij} = \dfrac{(i+2j)^2}{3}$. Then A' is:

(A) $\begin{bmatrix} 12 & \frac{25}{3} \\ \frac{16}{3} & 3 \end{bmatrix}$ **(B)** $\begin{bmatrix} 12 & \frac{16}{3} \\ \frac{25}{3} & 3 \end{bmatrix}$ **(C)** $\begin{bmatrix} 3 & \frac{25}{3} \\ \frac{16}{3} & 12 \end{bmatrix}$ **(D)** $\begin{bmatrix} 3 & \frac{16}{3} \\ \frac{25}{3} & 12 \end{bmatrix}$

1 Find each element.

$a_{11} = \dfrac{9}{3} = 3, \quad a_{12} = \dfrac{25}{3}, \quad a_{21} = \dfrac{16}{3}, \quad a_{22} = \dfrac{36}{3} = 12$

2 Write A.

$A = \begin{bmatrix} 3 & \frac{25}{3} \\ \frac{16}{3} & 12 \end{bmatrix}$

3 Interchange rows and columns to get A'. ✓ 1

$A' = \begin{bmatrix} 3 & \frac{16}{3} \\ \frac{25}{3} & 12 \end{bmatrix}$

4 Hence option (D).

Answer **(D)** $\begin{bmatrix} 3 & \frac{16}{3} \\ \frac{25}{3} & 12 \end{bmatrix}$

Watch out: Stopping at A (option C) and forgetting to take the transpose.

5 MCQ 1 mark

If A is a square matrix such that $A^2 = A$, then $(I + A)^3 - 7A$ is equal to:

(A) A **(B)** $I - A$ **(C)** I **(D)** $3A$

Formula / idea: A and I commute ($AI = IA = A$), so $(I + A)^3$ expands like $(1 + a)^3$.

1 From $A^2 = A$, get A^3.

$A^3 = A^2 \cdot A = A \cdot A = A^2 = A$

2 Expand $(I + A)^3$.

$(I + A)^3 = I + 3A + 3A^2 + A^3$

Allowed because $AI = IA$.

3 Put $A^2 = A$, $A^3 = A$.

$(I + A)^3 = I + 3A + 3A + A = I + 7A$

4 Subtract $7A$. ✓ 1

$(I + A)^3 - 7A = I + 7A - 7A = I$

5 Hence option (C).

Answer **(C)** I

6 **Assertion-Reason** 1 mark

Assertion (A): If A and B are two square matrices such that AB and BA are defined, then it is not necessary that $AB = BA$. **Reason (R):** The product of two diagonal matrices of the same order is commutative.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) **Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).**

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

1 Assertion: matrix multiplication is not commutative in general.
e.g. $A = \begin{bmatrix}1 & 1\\0 & 0\end{bmatrix}$, $B = \begin{bmatrix}1 & 0\\1 & 0\end{bmatrix}$: $AB = \begin{bmatrix}2 & 0\\0 & 0\end{bmatrix}$, $BA = \begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}$

2 So $AB \neq BA$ is possible. **A is true.**

3 Reason: diagonal matrices multiply entry by entry on the diagonal.
$\operatorname{diag}(a_1, a_2)\operatorname{diag}(b_1, b_2) = \operatorname{diag}(a_1b_1, a_2b_2) = \operatorname{diag}(b_1, b_2)\operatorname{diag}(a_1, a_2)$

4 So **R is true.**

5 R is about a special case where products commute; it does not explain A. ✓ 1

6 Hence option (B).

Answer **(B) Both A and R are true, but R is not the correct explanation of A.**

7 **Short answer** 2 marks

Find x and y if $2\begin{bmatrix}1 & 3\\0 & x\end{bmatrix} + \begin{bmatrix}y & 0\\1 & 2\end{bmatrix} = \begin{bmatrix}5 & 6\\1 & 8\end{bmatrix}$.

Formula / idea: Equal matrices have equal corresponding elements.

1 Multiply the first matrix by 2.
$\begin{bmatrix}2 & 6\\0 & 2x\end{bmatrix} + \begin{bmatrix}y & 0\\1 & 2\end{bmatrix} = \begin{bmatrix}5 & 6\\1 & 8\end{bmatrix}$

2	Add the matrices on the LHS. $\begin{bmatrix} 2+y & 6 \\ 1 & 2x+2 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 1 & 8 \end{bmatrix}$	✓ 1
3	Equate the $(1,1)$ elements. $2+y=5 \Rightarrow y=3$	✓ ½
4	Equate the $(2,2)$ elements. $2x+2=8 \Rightarrow x=3$	✓ ½
5	Hence $x=3$ and $y=3$.	

Answer $x=3,\ y=3$

Level 2

18 marks

8 **Short answer** 2 marks

Construct a 3×2 matrix $A=[a_{ij}]$ whose elements are given by $a_{ij}=\dfrac{1}{2}|i-3j|$.

Formula / idea: A 3×2 matrix has $i=1,2,3$ (rows) and $j=1,2$ (columns).

1	Row 1: put $i=1,\ j=1,2$. $a_{11}=\frac{1}{2}\lvert 1-3\rvert=1, \qquad a_{12}=\frac{1}{2}\lvert 1-6\rvert=\frac{5}{2}$	✓ ½
2	Row 2: put $i=2,\ j=1,2$. $a_{21}=\frac{1}{2}\lvert 2-3\rvert=\frac{1}{2}, \qquad a_{22}=\frac{1}{2}\lvert 2-6\rvert=2$	✓ ½
3	Row 3: put $i=3,\ j=1,2$. $a_{31}=\frac{1}{2}\lvert 3-3\rvert=0, \qquad a_{32}=\frac{1}{2}\lvert 3-6\rvert=\frac{3}{2}$	✓ ½
4	Write the required matrix. $A=\begin{bmatrix} 1 & \frac{5}{2} \\ \frac{1}{2} & 2 \\ 0 & \frac{3}{2} \end{bmatrix}$	✓ ½

Answer $A=\begin{bmatrix} 1 & \frac{5}{2} \\ \frac{1}{2} & 2 \\ 0 & \frac{3}{2} \end{bmatrix}$

Watch out: Dropping the modulus and writing negative entries like -1 instead of 1.

9 **Short answer** 3 marks

If $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, show that $A^2 - 2A - 3I = O$. Hence find A^{-1}.

1 Find $A^2 = A \cdot A$. ✓ 1

$$A^2 = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1+4 & 2+2 \\ 2+2 & 4+1 \end{bmatrix} = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$$

2 Substitute in $A^2 - 2A - 3I$. ✓ ½

$$\begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix} - \begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

3 Simplify. ✓ ½

$$= \begin{bmatrix} 5-2-3 & 4-4-0 \\ 4-4-0 & 5-2-3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$$

4 A^{-1} exists since $|A| \neq 0$.

$$|A| = 1 - 4 = -3 \neq 0$$

5 Multiply $A^2 - 2A - 3I = O$ by A^{-1} on both sides. ✓ ½

$$A^{-1}A^2 - 2A^{-1}A - 3A^{-1} = A^{-1}O \;\Rightarrow\; A - 2I - 3A^{-1} = O$$

6 Make A^{-1} the subject and simplify. ✓ ½

$$A^{-1} = \frac{1}{3}(A - 2I) = \frac{1}{3}\begin{bmatrix} 1-2 & 2 \\ 2 & 1-2 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix}$$

Answer $A^{-1} = \frac{1}{3}(A - 2I) = \frac{1}{3}\begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix}$

Watch out: Writing $A - 2 - 3A^{-1}$: the constant must stay a matrix, $2I$, not the number 2.

10 **Short answer** 3 marks

Express $A = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}$ as the sum of a symmetric and a skew-symmetric matrix.

Formula / idea: $A = P + Q$, where $P = \frac{1}{2}(A + A')$ is symmetric and $Q = \frac{1}{2}(A - A')$ is skew-symmetric.

1 Write A'. ✓ ½

$$A' = \begin{bmatrix} 3 & 1 \\ 5 & -1 \end{bmatrix}$$

2 Find $P = \frac{1}{2}(A + A')$. ✓ ½

$$P = \frac{1}{2}\begin{bmatrix} 6 & 6 \\ 6 & -2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & -1 \end{bmatrix}$$

3 Check P is symmetric. ✓ ½

$$P' = \begin{bmatrix} 3 & 3 \\ 3 & -1 \end{bmatrix} = P$$

4 Find $Q = \frac{1}{2}(A - A')$. ✓ ½

$$Q = \frac{1}{2}\begin{bmatrix} 0 & 4 \\ -4 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

5 Check Q is skew-symmetric. ✓ ½

$$Q' = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} = -Q$$

6 Add to verify and conclude. ✓ ½

$$P + Q = \begin{bmatrix} 3 & 3 \\ 3 & -1 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix} = A$$

Answer $A = \begin{bmatrix} 3 & 3 \\ 3 & -1 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$

Watch out: Skipping the checks $P' = P$ and $Q' = -Q$; examiners give marks for them.

11 Short answer — 2 marks

If $A = \begin{bmatrix} 8 & 0 \\ 4 & -2 \\ 3 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -2 \\ 4 & 2 \\ -5 & 1 \end{bmatrix}$, find the matrix X such that $2A + 3X = 5B$.

1 Make X the subject. ✓ ½

$$3X = 5B - 2A \Rightarrow X = \frac{1}{3}(5B - 2A)$$

2 Find $5B$ and $2A$. ✓ ½

$$5B = \begin{bmatrix} 10 & -10 \\ 20 & 10 \\ -25 & 5 \end{bmatrix}, \qquad 2A = \begin{bmatrix} 16 & 0 \\ 8 & -4 \\ 6 & 12 \end{bmatrix}$$

3 Subtract. ✓ ½

$$5B - 2A = \begin{bmatrix} 10 - 16 & -10 - 0 \\ 20 - 8 & 10 + 4 \\ -25 - 6 & 5 - 12 \end{bmatrix} = \begin{bmatrix} -6 & -10 \\ 12 & 14 \\ -31 & -7 \end{bmatrix}$$

4 Divide by 3. ✓ ½

$$X = \begin{bmatrix} -2 & -\frac{10}{3} \\ 4 & \frac{14}{3} \\ -\frac{31}{3} & -\frac{7}{3} \end{bmatrix}$$

Answer $X = \begin{bmatrix} -2 & -\frac{10}{3} \\ 4 & \frac{14}{3} \\ -\frac{31}{3} & -\frac{7}{3} \end{bmatrix}$

Watch out: Sign slip in $10 - (-4)$: the $(2, 2)$ entry is 14, not 6.

12 **Short answer** 3 marks

Find x if $\begin{bmatrix} x & -5 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ 4 \\ 1 \end{bmatrix} = O$.

Formula / idea: Multiply left to right: $(1 \times 3)(3 \times 3) = 1 \times 3$, then $(1 \times 3)(3 \times 1) = 1 \times 1$.

1 Multiply the first two matrices (row into columns).
$\begin{bmatrix} x + 0 - 2 & 0 - 10 + 0 & 2x - 5 - 3 \end{bmatrix}$

2 Simplify. ✓ 1
$= \begin{bmatrix} x - 2 & -10 & 2x - 8 \end{bmatrix}$

3 Multiply by the column matrix. ✓ ½
$\begin{bmatrix} x - 2 & -10 & 2x - 8 \end{bmatrix} \begin{bmatrix} x \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} x(x-2) - 40 + 2x - 8 \end{bmatrix}$

4 Simplify. ✓ ½
$= \begin{bmatrix} x^2 - 2x - 40 + 2x - 8 \end{bmatrix} = \begin{bmatrix} x^2 - 48 \end{bmatrix}$

5 Equate to the zero matrix $O = [0]$. ✓ ½
$x^2 - 48 = 0 \Rightarrow x^2 = 48$

6 Take square roots. ✓ ½
$x = \pm\sqrt{48} = \pm 4\sqrt{3}$

Answer $x = \pm 4\sqrt{3}$

Watch out: Giving only $x = 4\sqrt{3}$; both signs satisfy $x^2 = 48$.

13 **Short answer** 2 marks

If $A = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$, find $\alpha \left(0 < \alpha < \frac{\pi}{2}\right)$ such that $A + A' = I$.

1 Write A'. ✓ ½
$A' = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$

2 Find $A + A'$. ✓ ½
$A + A' = \begin{bmatrix} 2\cos\alpha & 0 \\ 0 & 2\cos\alpha \end{bmatrix}$

3 Equate to I. ✓ ½
$\begin{bmatrix} 2\cos\alpha & 0 \\ 0 & 2\cos\alpha \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow 2\cos\alpha = 1$

4 Solve in the given interval. ✓ ½

$\cos\alpha = \frac{1}{2} = \cos\frac{\pi}{3} \Rightarrow \alpha = \frac{\pi}{3}$

$\frac{\pi}{3}$ is the only value in $\left(0, \frac{\pi}{2}\right)$.

Answer $\alpha = \frac{\pi}{3}$

14 **Short answer** 3 marks

If $A = \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 2 & 1 \end{bmatrix}$, verify that $(AB)' = B'A'$.

1 Find AB (order 3×1 times 1×3 gives 3×3). ✓ 1

$AB = \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix} \begin{bmatrix} -1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 2 & 1 \\ 4 & -8 & -4 \\ -3 & 6 & 3 \end{bmatrix}$

2 Take the transpose: LHS. ✓ ½

$(AB)' = \begin{bmatrix} -1 & 4 & -3 \\ 2 & -8 & 6 \\ 1 & -4 & 3 \end{bmatrix}$

3 Write B' and A'. ✓ ½

$B' = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \quad A' = \begin{bmatrix} 1 & -4 & 3 \end{bmatrix}$

4 Find $B'A'$: RHS. ✓ ½

$B'A' = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & -4 & 3 \end{bmatrix} = \begin{bmatrix} -1 & 4 & -3 \\ 2 & -8 & 6 \\ 1 & -4 & 3 \end{bmatrix}$

5 LHS = RHS, so $(AB)' = B'A'$. Hence verified. ✓ ½

Answer $(AB)' = B'A' = \begin{bmatrix} -1 & 4 & -3 \\ 2 & -8 & 6 \\ 1 & -4 & 3 \end{bmatrix}$**, hence verified.**

Watch out: Computing $A'B'$ instead of $B'A'$; the order reverses.

Level 3

23 marks

15 **Long answer** 3 marks

If $A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$, find $A^2 - 5A + 4I$ and hence find a matrix X such that $A^2 - 5A + 4I + X = O$.

1 Find $A^2 = A \cdot A$ (row 1).

$R_1 : [4+0+1,\ 0+0-1,\ 2+0+0] = [5,\ -1,\ 2]$

2 Rows 2 and 3.

$R_2 : [4+2+3,\ 0+1-3,\ 2+3+0] = [9,\ -2,\ 5]$

$R_3 : [2-2+0,\ 0-1+0,\ 1-3+0] = [0,\ -1,\ -2]$

3 So ✓ 1

$$A^2 = \begin{bmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix}$$

4 Write $5A$ and $4I$. ✓ ½

$$5A = \begin{bmatrix} 10 & 0 & 5 \\ 10 & 5 & 15 \\ 5 & -5 & 0 \end{bmatrix}, \qquad 4I = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

5 Combine entry by entry.

$$A^2 - 5A + 4I = \begin{bmatrix} 5-10+4 & -1-0+0 & 2-5+0 \\ 9-10+0 & -2-5+4 & 5-15+0 \\ 0-5+0 & -1+5+0 & -2-0+4 \end{bmatrix}$$

6 Simplify. ✓ 1

$$A^2 - 5A + 4I = \begin{bmatrix} -1 & -1 & -3 \\ -1 & -3 & -10 \\ -5 & 4 & 2 \end{bmatrix}$$

7 Now $A^2 - 5A + 4I + X = O$ gives X. ✓ ½

$$X = -(A^2 - 5A + 4I) = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & 10 \\ 5 & -4 & -2 \end{bmatrix}$$

Answer $A^2 - 5A + 4I = \begin{bmatrix} -1 & -1 & -3 \\ -1 & -3 & -10 \\ -5 & 4 & 2 \end{bmatrix}$; $X = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & 10 \\ 5 & -4 & -2 \end{bmatrix}$

Watch out: Adding 4 to every entry instead of only the diagonal: $4I$ has 4 on the diagonal only.

16 **Long answer** 3 marks

A trust fund has ₹30,000 to be invested in two types of bonds paying 5% and 7% interest per annum. Using matrix multiplication, determine how to divide ₹30,000 among the two bonds so that the trust fund earns an annual interest of ₹1,800.

1 Let ₹x go in the first bond (5%).
$\text{Amount in second bond } (7\%) = ₹\,(30000 - x)$ ✓ ½

2 Write the interest as (investment row) × (rate column).
$$\begin{bmatrix} x & 30000 - x \end{bmatrix} \begin{bmatrix} \frac{5}{100} \\ \frac{7}{100} \end{bmatrix} = \begin{bmatrix} 1800 \end{bmatrix}$$ ✓ 1

3 Multiply.
$$\frac{5x}{100} + \frac{7(30000 - x)}{100} = 1800$$ ✓ ½

4 Multiply both sides by 100 and simplify.
$5x + 210000 - 7x = 180000 \;\Rightarrow\; 2x = 30000 \;\Rightarrow\; x = 15000$ ✓ ½

5 Second bond.
$30000 - 15000 = 15000$

6 Invest ₹15,000 in the 5% bond and ₹15,000 in the 7% bond. ✓ ½

Answer **₹15,000 in the 5% bond and ₹15,000 in the 7% bond.**

Watch out: Forgetting to convert percent to $\frac{5}{100}$, $\frac{7}{100}$ before multiplying.

17 **Long answer** 3 marks

If $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, show that $A^2 = 3A$ and hence find A^3.

1 Find A^2: each entry is $1 + 1 + 1$.
$$A^2 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{bmatrix}$$ ✓ 1

2 Take 3 common.
$$A^2 = 3 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = 3A$$ ✓ ½

3 Write $A^3 = A^2 \cdot A$ and use $A^2 = 3A$.
$A^3 = A^2 A = (3A)A = 3A^2$ ✓ ½

4 Use $A^2 = 3A$ again. ✓ ½
$A^3 = 3(3A) = 9A$

5 Write the matrix. ✓ ½
$A^3 = \begin{bmatrix} 9 & 9 & 9 \\ 9 & 9 & 9 \\ 9 & 9 & 9 \end{bmatrix}$

Answer $A^3 = 9A = \begin{bmatrix} 9 & 9 & 9 \\ 9 & 9 & 9 \\ 9 & 9 & 9 \end{bmatrix}$

18 **Long answer** 5 marks

If $A = \begin{bmatrix} 0 & -\tan\frac{\alpha}{2} \\ \tan\frac{\alpha}{2} & 0 \end{bmatrix}$ and I is the identity matrix of order 2, show that
$I + A = (I - A)\begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$.

Formula / idea: With $t = \tan\frac{\alpha}{2}$: $\cos\alpha = \frac{1-t^2}{1+t^2}$ and $\sin\alpha = \frac{2t}{1+t^2}$.

1 Let $t = \tan\frac{\alpha}{2}$. Then
$A = \begin{bmatrix} 0 & -t \\ t & 0 \end{bmatrix}$

2 Find the LHS. ✓ ½
$I + A = \begin{bmatrix} 1 & -t \\ t & 1 \end{bmatrix}$

3 Find $I - A$. ✓ ½
$I - A = \begin{bmatrix} 1 & t \\ -t & 1 \end{bmatrix}$

4 Write $\cos\alpha$, $\sin\alpha$ in terms of t. ✓ 1
$\cos\alpha = \frac{1-t^2}{1+t^2}, \qquad \sin\alpha = \frac{2t}{1+t^2}$

5 RHS: multiply $(I - A)$ by the rotation matrix. ✓ 1
$\begin{bmatrix} 1 & t \\ -t & 1 \end{bmatrix}\begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} = \begin{bmatrix} \cos\alpha + t\sin\alpha & -\sin\alpha + t\cos\alpha \\ -t\cos\alpha + \sin\alpha & t\sin\alpha + \cos\alpha \end{bmatrix}$

6 Simplify the $(1,1)$ entry; the $(2,2)$ entry is the same. ✓ ½
$\cos\alpha + t\sin\alpha = \frac{1-t^2}{1+t^2} + \frac{2t^2}{1+t^2} = \frac{1+t^2}{1+t^2} = 1$

7 Simplify the $(1,2)$ entry. ✓ ½
$-\sin\alpha + t\cos\alpha = \frac{-2t + t - t^3}{1+t^2} = \frac{-t(1+t^2)}{1+t^2} = -t$

8 Simplify the $(2,1)$ entry. ✓ ½

$$\sin\alpha - t\cos\alpha = \frac{2t - t + t^3}{1+t^2} = \frac{t(1+t^2)}{1+t^2} = t$$

9 So RHS equals LHS. ✓ ½

$$\text{RHS} = \begin{bmatrix} 1 & -t \\ t & 1 \end{bmatrix} = I + A = \text{LHS}$$

10 Hence $I + A = (I - A)\begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$ is proved.

Answer **RHS simplifies to** $\begin{bmatrix} 1 & -\tan\frac{\alpha}{2} \\ \tan\frac{\alpha}{2} & 1 \end{bmatrix} = I + A$**. Hence proved.**

Watch out: Writing $I - A$ with the same signs as A; subtracting flips the signs of both off-diagonal entries.

19 Long answer — 5 marks

If $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & -2 & 1 \\ 4 & 2 & 1 \end{bmatrix}$, show that $A^3 - 23A - 40I = O$.

1 Find $A^2 = A \cdot A$, row by row. ✓ ½

$R_1 : [1+6+12,\ \ 2-4+6,\ \ 3+2+3]$

$R_2 : [3-6+4,\ \ 6+4+2,\ \ 9-2+1]$

$R_3 : [4+6+4,\ \ 8-4+2,\ \ 12+2+1]$

2 So ✓ 1

$$A^2 = \begin{bmatrix} 19 & 4 & 8 \\ 1 & 12 & 8 \\ 14 & 6 & 15 \end{bmatrix}$$

3 Find $A^3 = A^2 \cdot A$, row by row. ✓ ½

$R_1 : [19+12+32,\ \ 38-8+16,\ \ 57+4+8]$

$R_2 : [1+36+32,\ \ 2-24+16,\ \ 3+12+8]$

$R_3 : [14+18+60,\ \ 28-12+30,\ \ 42+6+15]$

4 So ✓ 1

$$A^3 = \begin{bmatrix} 63 & 46 & 69 \\ 69 & -6 & 23 \\ 92 & 46 & 63 \end{bmatrix}$$

5 Write $23A$ and $40I$. ✓ 1

$$23A = \begin{bmatrix} 23 & 46 & 69 \\ 69 & -46 & 23 \\ 92 & 46 & 23 \end{bmatrix}, \qquad 40I = \begin{bmatrix} 40 & 0 & 0 \\ 0 & 40 & 0 \\ 0 & 0 & 40 \end{bmatrix}$$

6 Substitute in $A^3 - 23A - 40I$.

$$\begin{bmatrix} 63-23-40 & 46-46-0 & 69-69-0 \\ 69-69-0 & -6+46-40 & 23-23-0 \\ 92-92-0 & 46-46-0 & 63-23-40 \end{bmatrix}$$

✓ ½

7 Simplify.

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O$$

✓ ½

8 Hence $A^3 - 23A - 40I = O$ is proved.

Answer $A^3 = \begin{bmatrix} 63 & 46 & 69 \\ 69 & -6 & 23 \\ 92 & 46 & 63 \end{bmatrix}$ **and** $A^3 - 23A - 40I = O$**. Hence proved.**

Watch out: Sign error in the $(2, 2)$ entry: $-6 - (-46) - 40 = 0$.

20 Case study — 4 marks

Case Study: Three schools P, Q and R organised a fair to raise funds by selling hand-made fans, mats and plates made from recycled material at ₹25, ₹100 and ₹50 each respectively. School P sold 40 fans, 50 mats and 20 plates; School Q sold 25 fans, 40 mats and 30 plates; School R sold 35 fans, 50 mats and 40 plates.

(i) Write the sales of the three schools as a 3×3 matrix and the prices as a column matrix. 1 mark

(ii) Find the funds collected by School P. 1 mark

(iii) Using matrix multiplication, find the funds collected by each school and the total funds collected. 2 marks

(i) — 1 mark

1 Rows = schools P, Q, R; columns = fans, mats, plates.

$$S = \begin{bmatrix} 40 & 50 & 20 \\ 25 & 40 & 30 \\ 35 & 50 & 40 \end{bmatrix}$$

✓ ½

2 Prices of fan, mat, plate as a column matrix.

$$C = \begin{bmatrix} 25 \\ 100 \\ 50 \end{bmatrix}$$

✓ ½

Answer (i) $S = \begin{bmatrix} 40 & 50 & 20 \\ 25 & 40 & 30 \\ 35 & 50 & 40 \end{bmatrix}$, $C = \begin{bmatrix} 25 \\ 100 \\ 50 \end{bmatrix}$

(ii) 1 mark

1 Multiply P's row by the price column. ✓ ½

$$\begin{bmatrix}40 & 50 & 20\end{bmatrix}\begin{bmatrix}25\\100\\50\end{bmatrix} = 1000 + 5000 + 1000$$

2 Funds collected by School P. ✓ ½

$= ₹\,7000$

Answer (ii) ₹7,000

(iii) 2 marks

1 Funds = SC (order 3×3 times 3×1). ✓ ½

$$SC = \begin{bmatrix}40 & 50 & 20\\25 & 40 & 30\\35 & 50 & 40\end{bmatrix}\begin{bmatrix}25\\100\\50\end{bmatrix}$$

2 Multiply each row by the column. ✓ 1

$$= \begin{bmatrix}1000 + 5000 + 1000\\625 + 4000 + 1500\\875 + 5000 + 2000\end{bmatrix} = \begin{bmatrix}7000\\6125\\7875\end{bmatrix}$$

3 Add for the total.

$7000 + 6125 + 7875 = 21000$

4 P: ₹7,000, Q: ₹6,125, R: ₹7,875; total ₹21,000. ✓ ½

Answer (iii) P: ₹7,000; Q: ₹6,125; R: ₹7,875; total ₹21,000

Watch out: Writing the price matrix as a row; it must be a column so that SC is defined.